

Response to Reviewers

IMMI — CrabNet Hyperparameter Benchmark Data Descriptor

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(author list pending — see Q11)

September 25, 2026 [DRAFT — placeholder responses]

Dear Editor,

[**TODO**] Cover letter. Thank the editor and reviewers; state that the manuscript has been revised as a Data Descriptor; summarise the headline changes in 2–4 sentences. [see review-response-plan.md §E]

Summary of major changes (placeholder — confirm after Q1–Q14):

- [**TODO**] Scope stated explicitly as a hyperparameter-optimization benchmark. [T2, Q2]
- [**TODO**] Sum-to-one constraint reframed and precisely defined; alloy-physics claim removed. [T1, Q3]
- [**TODO**] Primers on CrabNet, Matbench and Ax + a researcher-workflow paragraph added. [T3, Q7]
- [**TODO**] Data dictionary + FAIR statement added. [T7, E1]
- [**TODO**] Objective trade-off / Pareto analysis added. [T8, R1.2]
- [**TODO**] Figures consolidated; resolution/tick fixes; Pareto figure added. [T6, Q8]
- [**TODO**] Multi-fidelity axes defined (and/or claim softened). [T4, Q9]
- [**TODO**] Noise-model justification / ablation. [T9, Q4]
- [**TODO**] Optimizer demonstration or documented downstream uses. [T10, Q5]
- [**TODO**] Limitations paragraph + expanded related work. [T11, Q10]
- [**TODO**] Data-accuracy corrections (repeat count; order-of-magnitude claim). [T13, Q12]
- [**TODO**] Funding Information field completed (NSF DMR-1651668). [T12, E6]

Editor

Provide a data dictionary and demonstrate FAIR compliance: IMMI asks that data supporting published articles meet FAIR principles and carry sufficient metadata to describe provenance and enable reuse.

Author response. [**TODO**] [T7, E1] Point to the new data-dictionary table and FAIR paragraph in *Data Records*; give the location.

Resolve the ambiguity in the sum-to-one constraint (R1 #1, R1 #3, R2 #3).

Author response. [**TODO**] [T1, E2, Q3] Summarise the reframe and the new marginal-distribution figure; give the location.

Establish materials-community value, and make the scope explicit.

Author response. [TODO] [T2, E3, Q2]

Add context for the tools used: a short paragraph each on CrabNet, Matbench (the experimental band gap task and why it was chosen), and the Ax platform; and sketch how a typical researcher currently uses these tools, and what this benchmark lets them do better.

Author response. [TODO] [T3, E4, Q7]

Figure 6 is supplied at a resolution that makes several labels and values illegible; please supply it as vector art or at print resolution. The logarithmic y-axis tick labelling in Figure 4 is difficult to read. Figures 1–5 are five separate single-panel histograms of similar type and could be consolidated into one multi-panel figure.

Author response. [TODO] [T6, E5, Q8] Note the consolidation of Figs 1–5, the vector Fig 6, the Fig 4 tick fix, and the added Pareto figure.

NSF DMR-1651668 is acknowledged in the text but the Funding Information field in the submission record is blank. Please complete it.

Author response. [TODO] [T12, E6, Q11] Confirm the field has been completed.

Reviewer 1

The manuscript states “applying a contrived constraint that the sum of all parameters must equal one” and argues that this mimics metal-alloy composition constraints. This requires proper justification, as the constraint is artificial, and its physical motivation is possibly overstated.

Author response. [TODO] [T1, R1.1, Q3]

The dataset contains: MAE, RMSE, runtime, model size as outputs. But it is not clear whether these exhibit meaningful trade-offs. Are there genuine Pareto fronts, and are some objectives strongly correlated?

Author response. [TODO] [T8, R1.2] Report the Spearman table and Pareto fronts (MAE $\leftrightarrow$ RMSE $\rho \approx 0.97$; accuracy $\leftrightarrow$ runtime $\rho \approx -0.6$; model size $\approx$ independent) and reference the new figure.

Is Sobol sampling equivalent to conventional statistical uniform random sampling? The authors should show whether the final 23-dimensional dataset actually has approximately uniform marginal distributions and how the imposed constraint affects them.

Author response. [TODO] [T1, R1.3, Q3] Reference the new marginal-distribution figure (raw Sobol vs post-constraint).

To my best understanding the paper is about ML hyperparameter optimization benchmark based on a materials dataset, rather than a materials-design benchmark in the usual sense. This distinction should be made very clear.

Author response. [TODO] [T2, R1.4, Q2]

Authors should justify “SHAP analysis as shown in Figure 6 analyzes the difference between the true experimental output and the expected output of the model”. SHAP values quantify the contribution of features to a model prediction relative to a reference/expected prediction. They do not directly measure the difference between “true experimental output” and expected output.

Author response. [TODO] [T5, R1.5, Q6] Correct the wording (attribution vs error); confirm the fix or the figure’s removal.

Reviewer 2

Reviewer 2’s assessment is a single paragraph; it is split below into its distinct points for a structured response.

[Novelty] The novelty and scientific depth of the contribution are limited; the work is essentially a data generation and storage exercise, and the manuscript reads more as a technical report describing a computational pipeline than a research article.

Author response. [TODO] [T11, R2.a, Q2, Q10] Note the Data Descriptor article type and the sharpened contribution framing.

[SHAP] The SHAP analysis in Figure 6 remains superficial and does not provide substantial new understanding beyond what is intuitively expected (e.g., more epochs and larger batches increase runtime).

Author response. [TODO] [T5, R2.b, Q6]

[Multi-fidelity] The claim of “multi-fidelity” is not convincingly substantiated; the multi-fidelity dimension is not clearly defined, demonstrated, or exploited in a manner that would distinguish this work from a standard hyperparameter sweep.

Author response. [TODO] [T4, R2.c, Q9]

[Constraint] The authors themselves describe the summation-to-one constraint as “contrived”; it does not reproduce the physics, chemistry, or thermodynamic realities of actual compositional optimization, so the claimed bridge to real-world materials optimization remains aspirational.

Author response. [TODO] [T1, R2.d, Q3]

[Noise model] The treatment of heteroskedastic noise through percentile ranks lacks rigorous validation that it meaningfully improves surrogate fidelity compared to simpler noise models; no quantitative comparison or ablation is provided.

Author response. [TODO] [T9, R2.e, Q4]

[Optimizer demo] The practical utility of the benchmark is not convincingly established; no demonstration is provided showing that existing adaptive design or Bayesian optimization algorithms perform differently or more informatively on this dataset compared to established benchmarks such as Olympus or the particle packing benchmark previously published by the same group.

Author response. [TODO] [T10, R2.f, Q5]

[Scope] The scope is quite narrow, being tied exclusively to one neural network architecture and one materials property prediction task.

Author response. [TODO] [T2, T11, R2.g, Q10]

[Self-assessment] The writing would benefit from more critical self-assessment of limitations, a more thorough discussion of related work, and a clearer articulation of what specific optimization challenges this dataset enables that cannot already be addressed by existing resources.

Author response. [TODO] [T11, R2.h, Q10]

Reviewer 3

This was a fine effort at a data-descriptor type article. A bit more context would have been appreciated: (1) what CrabNet is, and why anyone would care about it; (2) what Matbench is; (3) the Ax platform.

Author response. [TODO] [T3, R3.1–R3.3, Q7]

The readers would really benefit from getting an idea how a typical researcher might use these tools, and how the development of this new benchmark would improve the quality of that work (beyond what is offered in the concluding bulleted list).

Author response. [TODO] [T3, T2, R3.4, Q7] Add the status-quo → with-benchmark workflow paragraph.